

1.4a

The Brain

Neuroplasticity and Tools of Discovery

What is the role of Biology in Psychology?

Biological Psychologists use advanced technologies to study the links between biological: {Genetic, Nu, Hormonal} & psychological processes.

Biological Psychology Discoveries

- Neurons conduct electricity & "talk" to each other by sending chemical messengers across a synapse
- Experiences wire our adaptive brain
- Functions are localized
- We integrate info processed in different brain systems to construct experiences of sights, sounds, meanings, memories, pain & passion

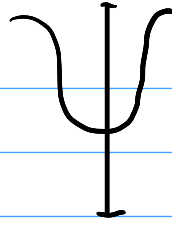
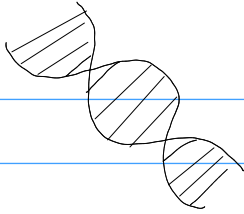
We are systems composed of subsystems, which are composed of their own subsystems.

Tiny cells organise to form our organs. Organs form larger systems (e.g. digestion, circulation, info processing). These systems are a part of an individual, who is part of a family, community, culture.

Cells & Organs & Body Systems & Individuals & Communities

∴ we are biopsychosocial systems.

To understand our behavior, we need to study how these biological, psychological, and social-cultural systems work, interact, & shape us. The biopsychosocial approach integrates these three levels of analysis.



Biological + Psychological + Socio cultural

- Genetic Predispositions
- Genetic Mutations
- Natural Selection of traits
- Epigenetics

- learned fears & expectations
- emotional responses
- cognitive processing
- perceptual interpretations

- presence of others
- cultural, societal, familial expectations
- peer & group influence
- compelling models (ex. media)

Behavior or Mental Process

How Does the Brain Change and Learn?

Brain is not only sculpted by genes but also by our lived experiences.

Brain is constantly changing, building new pathways

This change :: neuroplasticity.

Neuroplasticity is greatest in childhood, but it persists through life.

London's taxi drivers, after spending years learning & memorizing 25k street locations & connections develop a larger hippocampus.

Well practiced pianists have a larger-than-average auditory cortex.

Spending more 45min learning piano may grow motor learning related areas.

Neuroplasticity enables us to adapt to our very quickly changing world.

Personal Connection

For my Capstone, I plan to create something similar to Quizlet, but good. One thing I plan to do is use psychological concepts like neuroplasticity to design effective sorting algorithms.

How do Neuroscientists study the brain?

The mind is what the brain does

Scientists can selectively **lesion** small clusters of brain cells, **observing effect on brain function.**

Ex. damage to one part of hypothalamus made rat starve itself, damage to another made it overeat.

Scientists can also **stimulate various parts** to see a **reaction.**

Modern electrodes are precise enough to detect exactly where info goes in a rat's brain.

Optogenetics is a promising new tool that allows scientists to **control activity of individual neurons.**

Main Tools

Electroencephalogram

Amplified readout of brain waves.

Recorded through **shower-cap-like hat** filled with **electrodes** covered with a **conductive gel**

Researchers present stimulus repeatedly

Software filters out unrelated **brain activity**

Magnetoencephalography

Measures magnetic fields from brain's natural electrical activity.

Researchers use special rooms that cancel out other magnetic signals like Earth's magnetic field.

Participants sit under head coil

While they do activities, 10s or ks of neurons generate electrical pulses, which make magnetic fields.

Speed, strength (field) help understand how some tasks influence brain activity.

Computed Tomography Scan

Takes x-ray photos from different angles
photos combined by computer into composite representation of a slice of the brain's structure
can detect brain damage

Positron Emission Tomography

Depicts Brain Activity by showing each area's glucose consumption

Brain is 2% of body's mass, 20% of body's caloric consumption

Participant gets radioactive glucose - PET scan tracks the gamma rays released as a task is performed.

Magnetic Resonance Imaging

Head goes in Strong Magnetic field which
Aligns spinning atoms in brain molecules.
Then radio-wave pulse momentarily disorients atoms.
When atoms return to normal spin, they emit signals
that provide a detailed picture of soft tissues.

functional MRI

Blood goes where brain is active.
Compares multiple successive MRI scans to
see where blood flows to see where there is
activity.

functional near-infrared spectroscopy

Shines IR light on blood molecules
to identify blood flow to see brain activity.

fNIRS equipment is very portable
enabling researchers to study in difficult-
to-reach populations.

MCQ: 1d, 2c, 3b, 4b, 5c, 6a, 7c